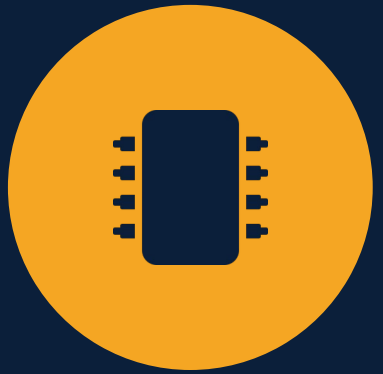
UNIT 3 · 9 hours

Diode V-I Characteristics and Equivalent Circuits



Diodes, Transistors and their Applications

Basic Electrical and Electronics Engineering (EEE103)

Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

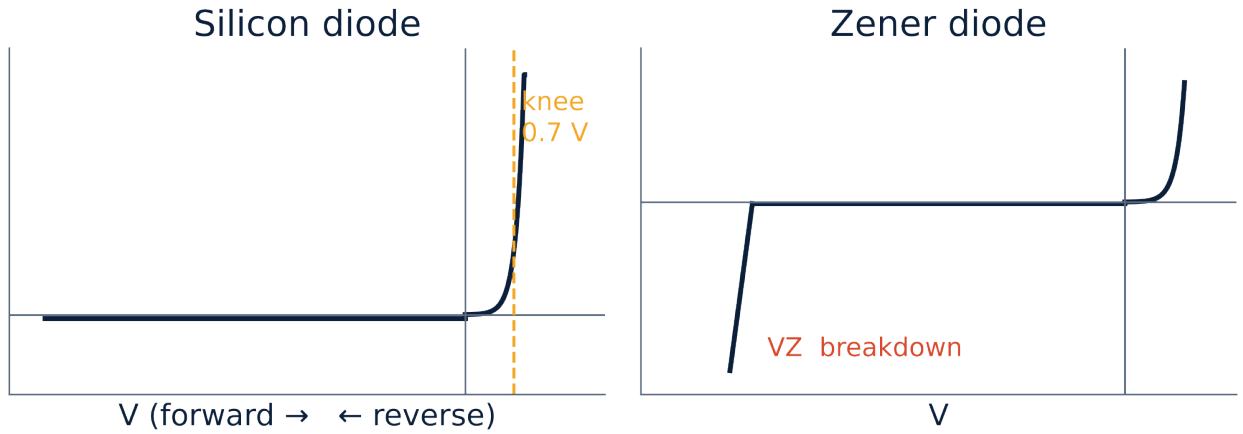
- Sketch and explain the V-I characteristic of a silicon diode
- Use the diode current equation
- Draw ideal, constant-voltage and piecewise-linear equivalent circuits
- Analyse simple diode circuits

Syllabus context

Semiconductor diode, diode equivalent circuits, half-wave rectifier, full-wave bridge rectifier. Zener diode and its use in voltage regulation.

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V-I characteristic of the silicon diode



- Forward: current negligible until the knee (≈ 0.7 V), then rises rapidly.
- Reverse: tiny saturation current I0 up to the breakdown voltage.
- Breakdown: large reverse current; the right plot shows the Zener case.

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Diode current equation

$$I = I_0 \left(e^{(V/\eta V_T)} - 1\right)$$

Shockley equation

$$V_T = kT/q \approx 26\ \text{mV}$$

at 300 K

- I_0 = reverse saturation current (nA–μA), η = 1 (Ge) to 2 (Si).
- Forward ($V \gg V_T$): $I \approx I_0 e^{(V/\eta V_T)}$, exponential.
- Reverse (V negative): $I \approx -I_0$.

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Silicon versus germanium

Property	Silicon	Germanium
Cut-in (knee) voltage	0.7 V	0.3 V
Reverse saturation current	nA	µA
Temperature stability	Good	Poor
Max. junction temperature	≈ 150 °C	≈ 75 °C

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Equivalent circuits of a forward-biased diode

Ideal

A switch with zero drop: V = 0 when ON, open when OFF.

Constant voltage

Ideal diode in series with a battery of 0.7 V (Si).

Piecewise linear

Battery V_γ in series with forward resistance rf: $V = V_\gamma + I_f r_f$.

Replace the diode by its equivalent circuit and use ordinary circuit analysis.

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Numerical: diode in series with a resistor

Problem A silicon diode is in series with 1 kΩ across a 12 V DC source (forward). Find the current using the constant-voltage model, and the power in the diode.

- $V_D = 0.7$ V
- $I = (12 - 0.7)/1000$
- $P_D = V_D \times I = 0.7 \times 11.3$ mA

Answer: I = 11.3 mA, PD = 7.9 mW

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Numerical: piecewise-linear model

Problem Diode with $V_\gamma = 0.7$ V and $r_f = 10\ \Omega$ in series with $R = 90\ \Omega$ across 10 V. Find the current.

- Total resistance = 90 + 10 = 100 Ω
- $I = (10 - 0.7)/100$

Answer: I = 93 mA

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Key takeaways

- Silicon knee ≈ 0.7 V; Germanium ≈ 0.3 V
- I = I0 (e^(V/ηVT) − 1)
- Equivalent circuits: ideal, constant-voltage, piecewise-linear
- Forward current limited by the series resistor

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Practice questions

Q1. Sketch the V-I curve of a silicon diode, marking knee and breakdown.

Q2. Find the current with 5 V, 470 Ω and a silicon diode.

Q3. Write the diode current equation and define each term.

Q4. What is VT at 27 °C?